

EXPERIMENT 10

Database with VIEW and INDEX

AIM

To create VIEW and INDEX on the Hospital database.

DESCRIPTION:

VIEW:

MySQL has supported database views since version 5+. In MySQL, almost features of views conform to the SQL: 2003 standard. MySQL processes query against the views in two ways:

1. In a first way, MySQL creates a temporary table based on the view definition
2. Statement and executes the incoming query on this temporary table.
3. In a second way, MySQL combines the incoming query with the query defined the view into one query and executes the combined query.

SYNTAX- VIEW:

CREATE [ALGORITHM = {MERGE | TEMPTABLE | UNDEFINED}]

VIEW [database_name].[view_name]

AS [SELECT statement]

INDEX:

A database index, or just index, helps **speed up the retrieval of data from tables**. When you query data from a table, first MySQL **checks if the indexes exist**, then MySQL uses the indexes to select exact physical corresponding rows of the table instead of scanning the whole table..

SYNTAX- INDEX:

CREATE

[UNIQUE|FULLTEXT|SPATIAL]

INDEXindex_name **USING** [BTREE |
HASH | RTREE]

ON table_name (column_name [(length)] [ASC | DESC],...)

PART A — QUESTIONS

1. Create a view named Patient_View using the Patient table that displays Patient_ID, Patient_Name and Disease.
2. Insert a new patient record through the Patient_View and display the contents of the view.
3. Delete a patient record through the Patient_View and display the contents of the view.
4. Create an index named idx_patient_name on the Patient_Name column of the Patient table and show it.

Output

1)

```
320 • CREATE VIEW Patient_View AS
321     SELECT Patient_ID, Patient_Name, Disease
322     FROM Patient;
323 • SELECT * FROM Patient_View;
```

Result Grid   Filter Rows: Export:  Wrap			
	Patient_ID	Patient_Name	Disease
+	P001	Arun Kumar	Fever
	P002	Priya	Diabetes
	P003	Rahul	Asthma
	P004	Sneha	Migraine

2)

```
326 • INSERT INTO Patient_View
327     VALUES ('P006', 'Karthik', 'Cold');
328 • SELECT * FROM Patient_View;
329
```

Result Grid			
Filter Rows:			
Export:  W			
	Patient_ID	Patient_Name	Disease
▶	P001	Arun Kumar	Fever
	P002	Priya	Diabetes
	P003	Rahul	Asthma
	P004	Sneha	Migraine
	P006	Karthik	Cold

3)

```
331 • DELETE FROM Patient_View
332     WHERE Patient_ID = 'P006';
333 • SELECT * FROM Patient_View;
334
```

Result Grid			
Filter Rows:			
Export: 			
	Patient_ID	Patient_Name	Disease
▶	P001	Arun Kumar	Fever
	P002	Priya	Diabetes
	P003	Rahul	Asthma
	P004	Sneha	Migraine

4)

```
336 • CREATE INDEX idx_patient_name
337     ON Patient(Patient_Name);
338 • SHOW INDEX FROM Patient;
339
340
341
342
343
```

Result Grid													
Filter Rows:													
Export:  Wrap Cell Content: 													
	Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment
▶	patient	0	PRIMARY	1	Patient_ID	A	4	NULL	NULL		BTREE		YES
	patient	0	Patient_Name	1	Patient_Name	A	4	NULL	NULL	YES	BTREE		YES
	patient	0	Patient_Name_2	1	Patient_Name	A	4	NULL	NULL	YES	BTREE		YES
	patient	1	idx_patient_name	1	Patient_Name	A	4	NULL	NULL	YES	BTREE		YES

RESULT:

The records from the tables are displayed using view and index on the given relation.

